

MATHEMATICS
Introduction to Modern Algebra II
Key to Quiz 1.

1) Prove that a finite domain is a division ring.

Solution. Let D be a finite domain. We must show that $1 \in D$ and every $0 \neq x \in D$ has an inverse. Consider the set $\{x, x^2, x^3, \dots\}$. Since D is finite, we must have $x^n = x^m$ for some integers $n < m$.

We claim that $x^{m-n} = 1$ i.e. that $x^{m-n}y = y$ for all $y \in D$. To see this, note that $x^n(x^{m-n}y - y) = x^m y - x^n y = 0$, but as D is a domain, we have $x^n \neq 0$ and hence $x^{m-n}y - y = 0$. This means that $x^{m-n}y = y$ and so $x^{m-n} = 1$.

Finally, note that the inverse of x is given by x^{m-n-1} .

2) Let I be an ideal in a ring R . Define addition and multiplication on R/I and show that the multiplication is well defined.

Solution. R/I is the set of left cosets $a + I$ where $a \in R$. We have $(a + I) + (b + I) = (a + b) + I$ and $(a + I)(b + I) = (ab) + I$. Suppose that $(a + I) = (a' + I)$ and $(b + I) = (b' + I)$ i.e. $a - a' \in I$ and $b - b' \in I$. In particular $a'b' \in I$, $(a - a')b' \in I$ and $a'(b - b') \in I$. Then

$$\begin{aligned}(a + I)(b + I) &= (a' + (a - a') + I)(b' + (b - b') + I) = \\ &= a'b' + (a - a')b' + a'(b - b') + I = a'b' + I.\end{aligned}$$